

Chunlin **Bian**

Target roles: Embedded Applications / Hardware–Software Systems / Embodied AI

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"Self-discipline and social commitment"

Profile

Tongji University Architecture Ph.D. candidate with hands-on experience in video-analysis software, smart-device UI development, InSAR monitoring, and full-stack AI applications. Implement earthquake-response algorithms in Windows/Linux and contribute UI design and code optimization to a smart lamp. Led a completed RMB 160,000 InSAR project, with related results spanning radar-reflector patents and monitoring software. Seeking hardware–software application and algorithm-engineering roles.

Technical Skills and Engineering Practice

- **Sensing applications:** Video-response algorithms; Windows/Linux development; smart-lamp UI and code optimization; InSAR and radar-reflector research.
- **Software:** Python, JavaScript/TypeScript, React, Node.js, REST APIs, Git; application interfaces and data-service integration.
- **AI and modeling:** RAG, LLM orchestration, multi-agent workflows; reinforcement learning and evacuation simulation; GIS/InSAR analysis.

Hardware–Software Project Experience

Video-Based Earthquake Detection and Response

Algorithm code and implementation | Windows / Linux; conventional cameras — *2026*

- **Implementation:** Implement earthquake-detection and response-analysis algorithms as software functions using footage from conventional cameras.
- **Engineering:** Develop in Windows and Linux environments, translating video-sensing research into application software.
- **Result:** Contributed to Earthquake Response Analysis Software Based on Surveillance Video V1.0; joint copyright holder, registered Sep 2026 (2026SR0931084).

Randeng Genie | Embedded Smart Desk Lamp

INTERCITY HACK, Ningbo | Visual/UI design and code optimization — *30 Oct–1 Nov 2026*

- **Contribution:** Designed the visual/UI layer and optimized code for posture reminders, facial-expression cues, and light/voice feedback.
- **System context:** Collaborated on the application layer around the team's proposed camera, five-state FSM, WS2812 lighting, and offline voice architecture.
- **Delivery:** Contributed to the team showcase at INTERCITY HACK, Ningbo, part of the Hangzhou–Ningbo–Wenzhou hackathon; project video available online.

InSAR Monitoring and Radar-Reflector Research

Project lead | Monitoring research, hardware-related patents, and software — *2018–2021*

- **Project leadership:** Led the completed Xichang InSAR subsidence-monitoring project funded by Liangshan Science and Technology Bureau (18YYJS0107; RMB 160,000).
- **Hardware-related results:** Contributed to radar-reflector research yielding two utility-model patents: a miniaturized bidirectional planar corner reflector and a rocker-arm angle bracket.
- **Software:** Developed research software for 3D terrain reconstruction, DInSAR error/decorrelation analysis, time-series deformation monitoring, and image interpretation; four copyright registrations.

Software Development Projects

StockMate AI

Independent full-stack developer | AI investment research — *2024–Present*

- **Full-stack delivery:** Independently built frontend interactions, backend services, financial-data APIs, and multi-model orchestration for an AI research product serving older investors.
- **AI engineering:** Connected specialist agents for macro, financial, technical, sentiment, and risk analysis with RAG retrieval and structured report generation.
- **Features:** Market monitoring, financial Q&A, and strategy simulation; advanced to the national second round through the 2026 Tongji AI Hackathon Tour. Project showcase.

Research and Algorithm Background

Seismic Resilience and Multi-Agent Evacuation

Tongji University | Doctoral research — *2021–Present*

- Investigate multi-agent reinforcement learning versus classical models for hospital evacuation under earthquakes; first-author paper in Buildings (2026).
- Doctoral research combines GEM seismic-hazard and GHSL population data, GIS, AHP–entropy weighting, and recovery curves to assess urban seismic resilience.
- Incorporate spatiotemporal population exposure and multi-agent evacuation simulations to study interactions among built environments, human behavior, and urban-renewal interventions.

Professional Experience

Xichang University | Former Senior Lecturer

GIS, parametric design, and research software — *Joined Feb 2016 (former)*

- Taught GIS, parametric design, urban planning, and landscape architecture; led an InSAR subsidence-monitoring project and contributed to deformation-monitoring research.
- Contributed to an InSAR/digital-earth laboratory, connecting spatial analysis and monitoring research with software development.

Yiren Architecture and Landscape Design Co., Ltd.

Founder — *Jan 2018–Present*

- Led rural-revitalization, forest-park, and heritage-area planning projects, coordinating requirements, design iterations, spatial analysis, and delivery.

Panzhihua University | Lecturer in Architecture

Architecture, urban design, and landscape architecture — *Aug 2010–Feb 2016*

- Taught courses, supervised graduation projects, and contributed to industry collaborations and local architectural research.

Education

- **Tongji University** — Ph.D. candidate, Architecture, Sep 2021–Present; seismic resilience, multi-agent simulation, and urban renewal.
- **Massey University** — Visiting Scholar, 2022; CSC-funded research in digital spatial analysis and landscape ecology.
- **Peking University** — Master's degree, Landscape Architecture, 2008–2010.
- **Sichuan Agricultural University** — Bachelor's degree, Landscape Architecture, 2003–2007.

Selected Results and Recognition

- **Hongmeng team lead; Third Prize, final round** — 2026 China (Guangxi)–ASEAN AI + Emergency Management Innovation Competition, Track 5 (Universities and ASEAN), Sep 2026; AI-driven urban seismic-resilience sensing and analysis platform.
- **Joint software copyright holder** — Surveillance-video earthquake response analysis (2026SR0931084); City Radar and Data Technology Platform V1.0 (2026SR0999998).
- **First-author paper** — Bian C, Guo Y, Meng G, et al. Modeling seismic resilience and hospital evacuation: A comparative analysis of multi-agent reinforcement learning and classical evacuation models. *Buildings*, 2026, 16(8): 1538.
- **First-author paper, accepted** — Bian C, Meng G, Hu B, Yang W. The Construction of "Republican-Era Style" in the Renewal of Yuantian Street, Huili Ancient City: Contemporary Translation of Modern Historical Imagery. *Housing Science* (in Chinese); accepted 29 Oct 2026, provisionally scheduled for issue 12, 2026.
- **First-author conference paper, planned publication** — Bian C, Meng G, Yang J. Structural Transformation and Functional Reconstruction in a Modern Historical Context: A Study of the Bell and Drum Tower in Huili Ancient City, Sichuan. Listed in the contents of Research and Conservation of Modern Chinese Architecture, Vol. 14; planned with Tianjin University Press (in Chinese).
- **Young Geographers Award — Best Paper**, 15th Korea–China–Japan Joint Conference on Geography, 2023.